

APS110 – Lecture 19

See hand written notes for drawings.

Polymers

- Mer units → repeating units that make up the structure of polymers
 - 'n' denotes the number of repeating mer units in a particular polymer
- Generally, formal naming is not used, in lieu of a simpler name
- PVC – polyvinyl chlorines
 - Chlorine in the mer unit increases the strength
 - Chlorine is quite electronegative
 - Introduces negative charge, which strengthens the interaction from one molecule to another
- Teflon® – polytetrafluoro ethylene
 - Fluorine is a rather large, electronegative atom → it protects the carbon backbone of the molecule
- By carefully choosing the chemistry of these mer units, you can greatly influence the physical properties of the resulting polymers
- Stress-strain curve
 - How is that the polymer can continue to support load after the onset of necking?
- Polymer molecules have interactions with each other, holding them together (non-primary bonds, ex. dipole interactions)
- **Chain orientation**
 - 'Chain', when discussing polymer chemistry, can be used synonymously with 'molecule'
 - Strong bonds, which are holding the molecule together (covalent bonds), orient to the loading axis
 - This means that these strong bonds are caused to support more load
 - This makes the material much stronger!
- As the polymer necks under plastic deformation, the necked polymer becomes very strong (as chain orientation occurs)
 - Then the regions around the neck are comparatively weaker and so they will start to deform and neck, which will then cause these regions to become very strong and the molecules align with the loading axis
- Ex. stretching a plastic bag
 - Imagine carrying a heavy load in a plastic shopping bag, and the handles of the bag will start to stretch and become thin and cut into your hand
 - The molecules of the bag orient as the bag stretches, and necking occurs
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 - Optical change → with a clear bag you can usually observe this, where as you continue to stretch the section of deformed bag becomes **opaque**
 - This is because the alignment of the molecules creates a sort of crystallinity in the structure, which light will not be able to penetrate and so you get opacity

- Some polymers, through carefully processing or careful cooling, can have some regions of crystal structure
 - o Crystal structure just involves the polymer folded back on itself over and over
 - o Microstructure of many polymers involves regions of crystal structure amongst regions of amorphous structure
 - o Light passing through these microstructures will refract and change directions as it passes through these different regions
 - Light can't go straight through a semi-crystalline polymer
 - The polymer will thus be translucent or even opaque
- Polymers may be semi-crystalline
 - o They can't be completely crystalline, but they also can't be strictly amorphous either